E3SM QuickCompare 1.8.3 Release (June 22, 2026)

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BER-ASCR SciDAC Partnership (PAESCAL), SciDAC Institute (RAPIDS)

Scientific Achievement

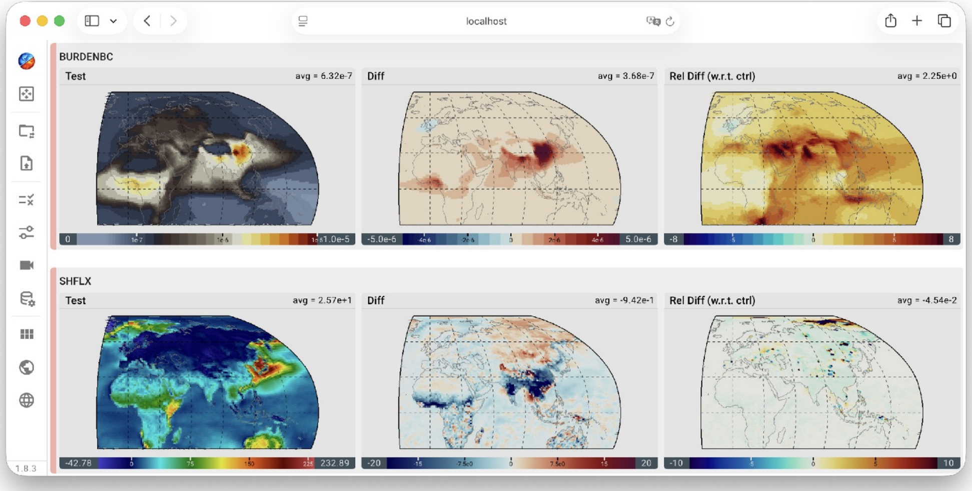
Developed an interactive visualization tool, E3SM QuickCompare, to facilitate quantitative comparison of sensitivity experiments and simulation ensembles.

Significance and Impact

- The development of complex simulation tools like the Energy Exascale Earth System Model (E3SM) often requires analysis of numerous variables across many sensitivity experiments. Intuitive data exploration enabled by QuickCompare can substantially speed up model developers' workflow for quantitative analysis.
- QuickCompare is the second tool in the QuickView family. The development further strengthened a deep collaboration between computer scientists and physical scientists. It also refined a strategy for providing advanced visual analytics tools for the development of E3SM and similar Earth system models. The development of subsequent tools is expected to become increasingly more efficient.

Technical Approach

- QuickCompare inherits an infrastructure from QuickView, using Python and trame to create a simple and intuitive User Interface (UI) to ParaView while providing tailored support for simulation comparison.
- Subsequent E3SM-specific applications are expected to also inherit the infrastructure — as well as further improve it when new capabilities are introduced.



A screenshot of E3SM QuickCompare, a lightweight interactive analysis tool tailored to the need of quantitative comparison of sensitivity experiments and simulation ensembles in the development of Earth system models like DOE's E3SM.

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Publication(s) for this work:
Code repository: https://github.com/Kitware/E3SMQuickCompare
Software website: https://kitware.github.io/QuickView/guides/quickcompare/